

Verena Bellscheidt

EDUCATION

Massachusetts Institute of Technology

2025 - present

Ph. D. Physics (GPA: 5.0/5.0)

Advisor: David Kaiser

Technical University of Munich

2021 - 2025

B. Sc. Physics (passed with distinction)

Thesis: *Neural Network Parametrisation of Generalised Wilson Loops in Lattice QCD*

Advisor: Nora Brambilla

INTERNSHIPS

Nicolaus Copernicus Astronomical Center

06/2023 - 10/2023

Undergraduate research internship

Project title: *Chemical abundances in a planet-host star of the Praesepe open cluster*

Advisor: Rodolfo Smiljanic

Max Planck Institute for Astrophysics

06/2022 - 10/2022

Undergraduate research internship

Project title: *Galaxy-IGM correlations in the Thesan simulations*

Advisor: Enrico Garaldi

PUBLICATIONS [[Inspire HEP](#),

- [1] **V. Bellscheidt**, N. Brambilla, A. S. Kronfeld, J. Mayer-Steudte. “Wilson loops with neural networks”. In: *Phys. Rev. D* (2026). DOI: [10.1103/qjpp-qw7x](#).
- [2] E. Garaldi, **V. Bellscheidt**, A. Smith, R. Kannan. “The galaxy-IGM connection in THESAN: the physics connecting the IGM Lyman- α opacity and galaxy density in the reionization epoch”. In: *The Open Journal of Astrophysics* (2025). DOI: [10.33232/001c.143245](#).
- [3] E. Garaldi, **V. Bellscheidt**, A. Smith, R. Kannan. “The galaxy-IGM connection in THESAN: observability and information content of the galaxy-Lyman- α cross-correlation at $z \geq 6$ ”. In: *The Open Journal of Astrophysics* (2025). DOI: [10.33232/001c.151666](#).

HONORS AND AWARDS

Zonta Quaboag Valley STEM award	Annually awarded to a female STEM researcher (\$2000)	2026
First-year fellowship	from the MIT department of physics, covering tuition and salary for one year	2025
Female physicist of the week	of the German Physical Society	2025
DAAD RISE worldwide scholarship	Full funding for a research internship abroad (€1984)	2023
Deutschlandstipendium	Germany’s national scholarship (€3600)	2021

TEACHING

MIT Teaching+Learning Lab

2025 - present

Grad Teaching Development Tracks

I am voluntarily completing the [Grad Teaching Development Tracks](#) of MIT's Teaching+Learning Lab in order to refine my teaching skills. Each track consists of 2-3 workshops in addition to pre- and post-workshop assignments. I have so far completed the tracks on subject design and inclusive teaching.

Technical University of Munich

2023 - 2025

Teaching assistant

Winter 2024/25: Introduction to Scientific Programming (IN8008) - *Led Ø1.5 tutorials per week*

Summer 2024: Theoretical Physics 1: Classical Mechanics (PH0005) - *Led weekly tutorials, graded homework& exams*

Winter 2023/24: Theoretical Physics 2: Electrodynamics (PH0006) - *Led weekly tutorials, graded homework& exams*

COMMUNITY SERVICE

MSRP admissions committee member

01/2026 - 02/2026

I helped select candidates for the MIT Summer Research Program (MSRP), a program which prepares promising students for graduate school, especially those with limited access to research. I read and evaluated applications, discussed them with the rest of the committee, and voted on the selection of candidates.

SCHOOLS & WORKSHOPS

YCIU Workshop on Primordial Black Holes Workshop at the Yale Center for the Invisible Universe 2026

Dark Matter and the Cosmos Summer school by TU Munich, FAU, and the University of Stuttgart 2022

SKILLS

Programming: Python, C++, Bash, Mathematica

Languages: English (C2), German (native speaker)